

Week11-AED-EMO-RMarkdown

Duong Hien Chi Kien V202401461

Tran Anh Chuong V202401566

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr    1.5.2
## v ggplot2    4.0.0      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
#install.packages('WDI')
library(WDI)
```

```
## Warning: package 'WDI' was built under R version 4.5.3
```

```
#install.packages('ggrepel')
library(ggrepel)
```

```
## Warning: package 'ggrepel' was built under R version 4.5.3
```

```
#install.packages('ggtext')
library(ggtext)
```

```
## Warning: package 'ggtext' was built under R version 4.5.3
```

```
indicators <- c("SP.POP.TOTL", "EN.GHG.CO2.PC.CE.AR5", "NY.GDP.PCAP.KD")

# Fetch the data
wdi_co2_raw <- WDI(country = "all", indicators, extra = TRUE,
                  start = 1995, end = 2023)
```

```
wdi_clean <- wdi_co2_raw |>
  filter(region != "Aggregates") |>
  select(iso2c, iso3c, country, year,
         population = SP.POP.TOTL,
         co2_emissions = EN.GHG.CO2.PC.CE.AR5, # Update the column name here!
         gdp_per_cap = NY.GDP.PCAP.KD,
         region, income
  ) |>
  filter(population > 200000)
```

```
co2_rankings <- wdi_clean |>
  # Get rid of all the rows that have missing values in co2_emissions
  drop_na(co2_emissions) |>
  # Look at each year individually and rank countries based on their emissions that year
  mutate(
    ranking = rank(co2_emissions),
    .by = year
  )
```

Task 1: Prepare data in wide format

```
# YOUR CODE HERE
co2_rankings_wide <- co2_rankings |>
  select(iso3c, country, year, region, income, ranking) |>
  pivot_wider(names_from = year, values_from = ranking,
print(head(co2_rankings_wide, 5))
```

```
## # A tibble: 5 x 33
##   iso3c country region income rank_1996 rank_2000 rank_2001 rank_2004 rank_2012
##   <chr> <chr>   <chr>   <chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 AFG  Afghani~ Middl~ Low i~      14         4         4         2         29
## 2 ALB  Albania Europ~ Upper~      50        64        67        74        67
## 3 DZA  Algeria Middl~ Upper~      93        93        92        95       101
## 4 AGO  Angola  Sub-S~ Lower~      64        63        62        59        59
## 5 ARG  Argenti~ Latin~ Upper~     103       109       105       107       113
## # i 24 more variables: rank_2013 <dbl>, rank_2003 <dbl>, rank_2011 <dbl>,
## #   rank_2007 <dbl>, rank_2008 <dbl>, rank_2009 <dbl>, rank_2006 <dbl>,
## #   rank_1998 <dbl>, rank_1999 <dbl>, rank_2010 <dbl>, rank_1997 <dbl>,
```

```
## #   rank_2023 <dbl>, rank_2014 <dbl>, rank_2015 <dbl>, rank_2002 <dbl>,
## #   rank_2005 <dbl>, rank_2018 <dbl>, rank_2019 <dbl>, rank_2016 <dbl>,
## #   rank_1995 <dbl>, rank_2022 <dbl>, rank_2020 <dbl>, rank_2021 <dbl>,
## #   rank_2017 <dbl>
```

```
print(colnames(co2_rankings_wide))
```

```
## [1] "iso3c"      "country"    "region"     "income"     "rank_1996"  "rank_2000"
## [7] "rank_2001"  "rank_2004"  "rank_2012"  "rank_2013"  "rank_2003"  "rank_2011"
## [13] "rank_2007"  "rank_2008"  "rank_2009"  "rank_2006"  "rank_1998"  "rank_1999"
## [19] "rank_2010"  "rank_1997"  "rank_2023"  "rank_2014"  "rank_2015"  "rank_2002"
## [25] "rank_2005"  "rank_2018"  "rank_2019"  "rank_2016"  "rank_1995"  "rank_2022"
## [31] "rank_2020"  "rank_2021"  "rank_2017"
```

Task 2: Data wrangling

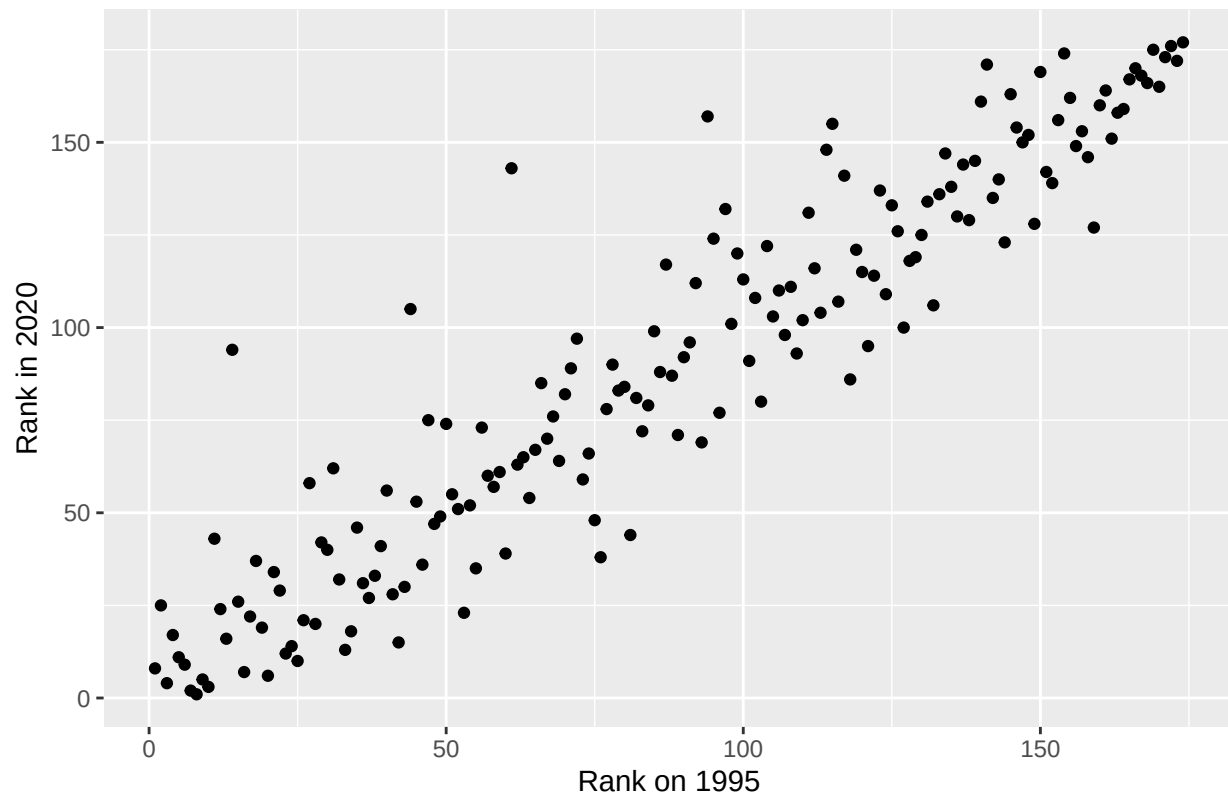
```
# YOUR CODE HERE
co2_rankings_draw <- co2_rankings_wide |>
  mutate(rank_diff = rank_2020 - rank_1995) |>
  drop_na(rank_diff) |>
  mutate(significant_diff = case_when(
    rank_diff <= -30 ~ "Significant decrease",
    rank_diff >= 30 ~ "Significant increase",
    .default = "Insignificant change"
  )) |>
  arrange(rank_diff)
```

Task 3: Scatter plot for changes in CO2 emission rankings between 1995 and 2020

```
# YOUR CODE HERE

ggplot(data = co2_rankings_draw, aes(x = rank_1995, y = rank_2020))+
  geom_point()+
  labs(
    title = "Changes in CO2 emission ranking between 1995 and 2020",
    x = "Rank on 1995", y = "Rank in 2020"
  )+
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    legend.position = "right"
  )
```

Changes in CO2 emission ranking between 1995 and 2020

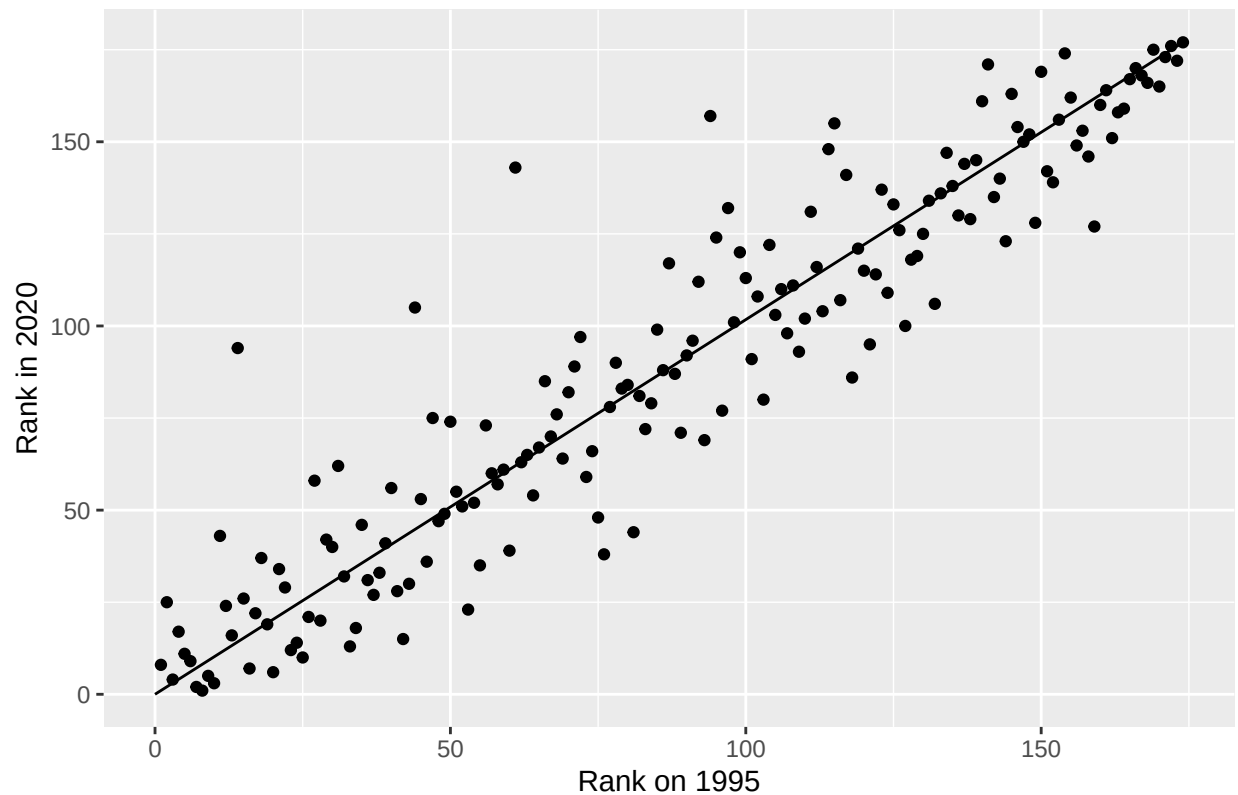


Task 4: Lazy way to show change in rank

```
# YOUR CODE HERE

ggplot(data = co2_rankings_draw, aes(x = rank_1995, y = rank_2020))+
  annotate(geom = "segment", x = 0, xend = max(co2_rankings_draw$rank_1995), y = 0, yend = max(co2_rankings_draw$rank_2020))+
  geom_point()+
  labs(
    title = "Changes in CO2 emission ranking between 1995 and 2020",
    x = "Rank on 1995", y = "Rank in 2020")+
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    legend.position = "right"
  )
```

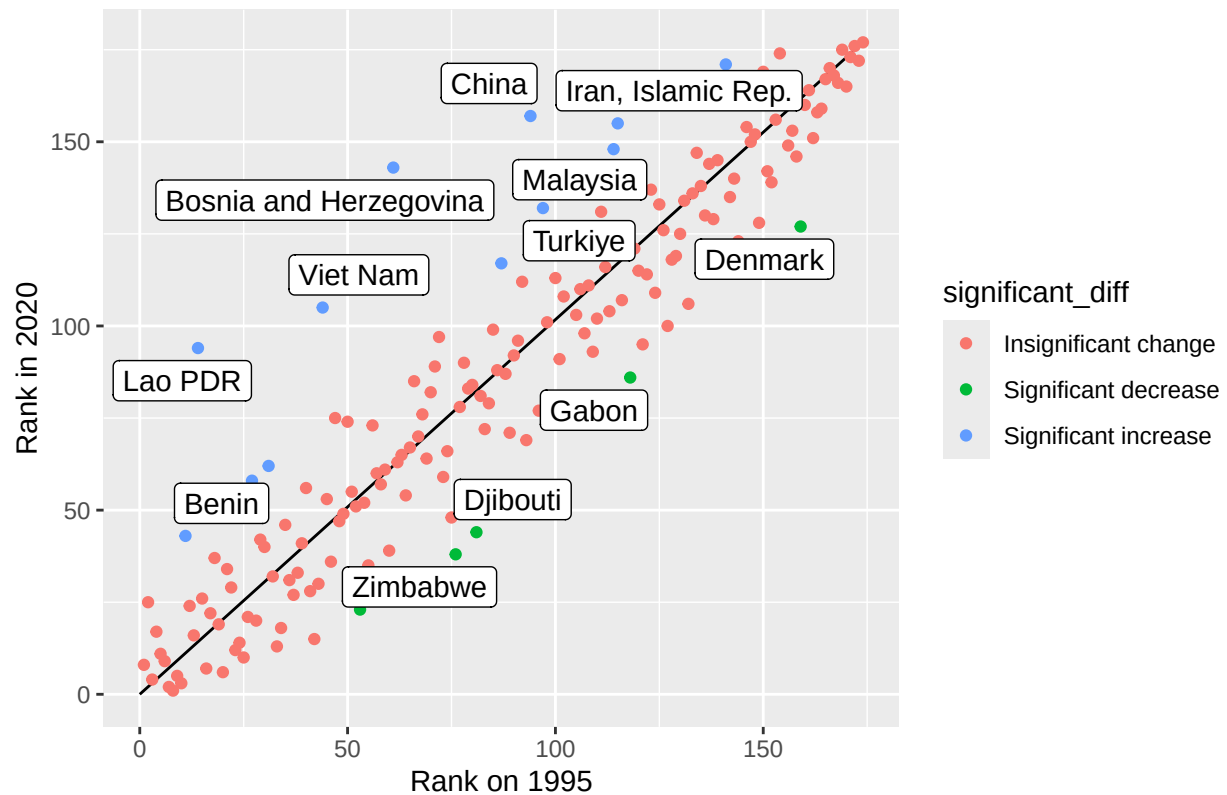
Changes in CO2 emission ranking between 1995 and 2020



Task 5: Highlight significant countries

```
# YOUR CODE HERE
co2_rankings_draw <- co2_rankings_draw |> mutate(sig_change = if_else(abs(rank_diff)>= 32, TRUE, FALSE))
ggplot(data = co2_rankings_draw, aes(x = rank_1995, y = rank_2020))+
  annotate(geom = "segment", x = 0, xend = max(co2_rankings_draw$rank_1995), y = 0,
    yend = max(co2_rankings_draw$rank_2020))+
  geom_point(aes(color = significant_diff))+
  geom_label_repel(
    data = filter(co2_rankings_draw, sig_change == TRUE),
    aes(label = country))+
  labs(
    title = "Changes in CO2 emission ranking between 1995 and 2020",
    x = "Rank on 1995", y = "Rank in 2020")+
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    legend.position = "right"
  )
```

Changes in CO2 emission ranking between 1995 and 2020

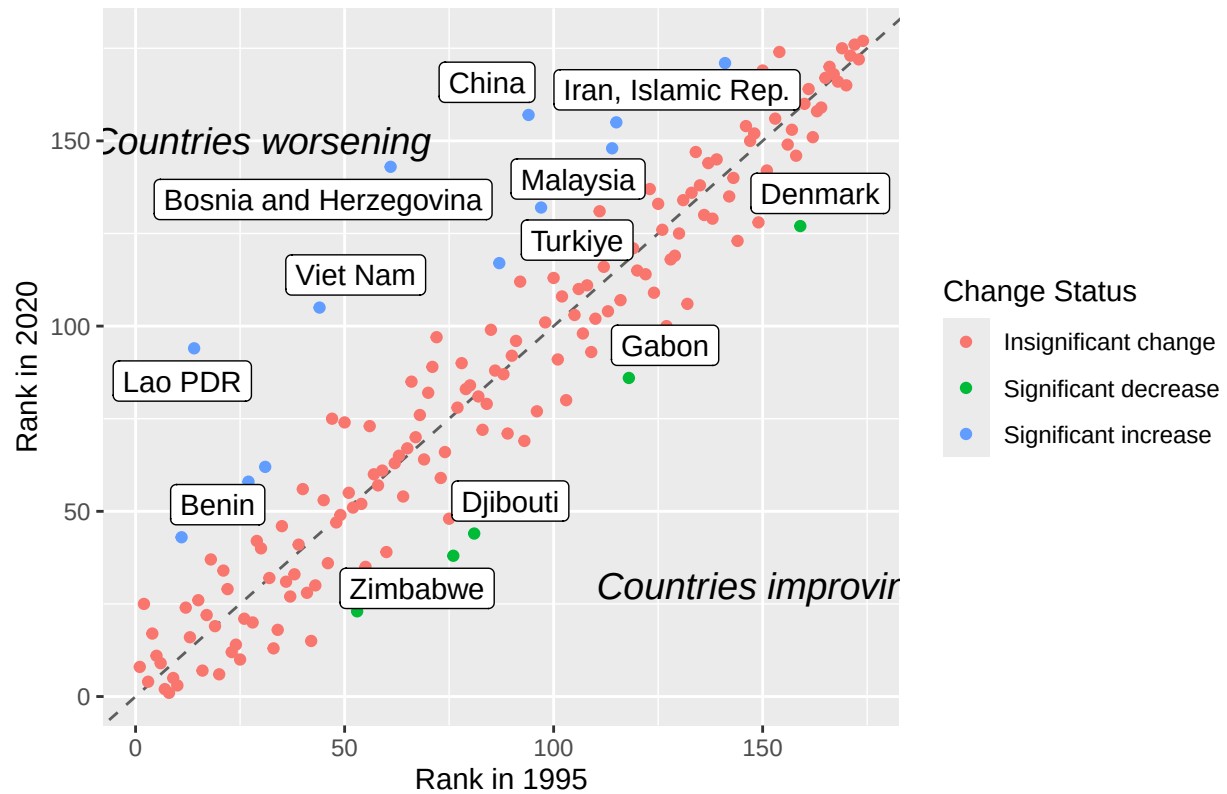


Task 6: Additional text annotations

YOUR CODE HERE

```
ggplot(data = co2_rankings_draw, aes(x = rank_1995, y = rank_2020)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", alpha = 0.6) +
  geom_point(aes(color = significant_diff)) +
  geom_label_repel(
    data = filter(co2_rankings_draw, sig_change == TRUE),
    aes(label = country),
    max.overlaps = 20
  ) +
  annotate(geom = "text", x = 150, y = 30, label = "Countries improving", size = 5, fontface = "italic") +
  annotate(geom = "text", x = 30, y = 150, label = "Countries worsening", size = 5, fontface = "italic") +
  labs(
    title = "Changes in CO2 emission ranking between 1995 and 2020",
    x = "Rank in 1995",
    y = "Rank in 2020",
    color = "Change Status"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "right"
  )
```

Changes in CO2 emission ranking between 1995 and 2020



Task 7: Using colors to redirect attention

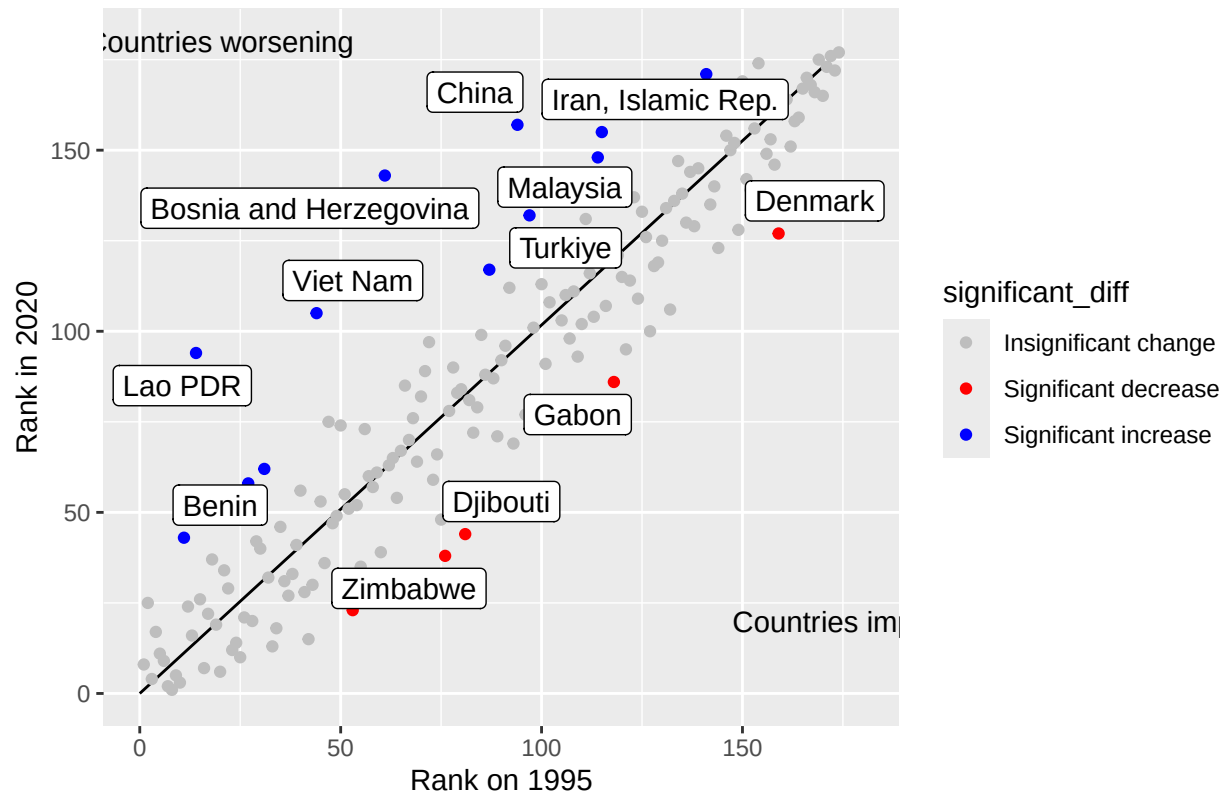
```
# YOUR CODE HERE
ggplot(data = co2_rankings_draw, aes(x = rank_1995, y = rank_2020))+
  annotate(geom = "segment", x = 0, xend = max(co2_rankings_draw$rank_1995), y = 0,
    yend = max(co2_rankings_draw$rank_2020))+
  geom_point(aes(color = significant_diff))+
  geom_label_repel(
    data = filter(co2_rankings_draw, sig_change == TRUE),
    aes(label = country)
  )+
  annotate(
    geom = "text", x = 180, y = 20, label = "Countries improving"
  )+
  annotate(
    geom = "text", x = 20, y = 180, label = "Countries worsening"
  )+
  labs(
    title = "Changes in CO2 emission ranking between 1995 and 2020",
    x = "Rank on 1995", y = "Rank in 2020"
  )+
```

```

theme(
  plot.title = element_text(hjust = 0.5),
  plot.subtitle = element_text(hjust = 0.5),
  legend.position = "right"
) +
  scale_color_manual(values = c("grey", "red", "blue"))

```

Changes in CO2 emission ranking between 1995 and 2020



Task 8: More geometric annotations

```

# Finding the max rank to calculate the top/bottom 25 accurately
max_rank <- max(c(co2_rankings_draw$rank_1995, co2_rankings_draw$rank_2020), na.rm = TRUE)

ggplot(data = co2_rankings_draw, aes(x = rank_1995, y = rank_2020)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", alpha = 0.6) +
  geom_point(aes(color = significant_diff)) +
  geom_label_repel(
    data = filter(co2_rankings_draw, sig_change == TRUE),
    aes(label = country),
    max.overlaps = 20
  ) +
  scale_color_manual(values = c(
    "Significant increase" = "red",

```



```

    "Significant decrease" = "blue",
    "Insignificant change" = "grey80"
  )) +

  # Text for halves
  annotate(geom = "text", x = 150, y = 30, label = "Countries improving", size = 5, fontface = "italic") +
  annotate(geom = "text", x = 30, y = 150, label = "Countries worsening", size = 5, fontface = "italic") +

  # Lowest Emitters Annotation (Top 25 rank, since lower rank = lower emissions)
  annotate('rect', xmin = 1, xmax = 25, ymin = 1, ymax = 25, fill = "green", alpha = 0.2) +
  annotate('text', x = 40, y = 6, color = "darkgreen", label = "Lowest emitters", hjust = 0, fontface = "italic") +
  annotate('segment', x = 38, xend = 26, y = 6, yend = 12, color = "darkgreen",
           arrow = arrow(type = "closed", length = unit(0.1, "inches"))) +

  # Highest Emitters Annotation (Bottom 25 rank)
  annotate('rect', xmin = max_rank - 25, xmax = max_rank, ymin = max_rank - 25, ymax = max_rank, fill = "darkorange", alpha = 0.2) +
  annotate('text', x = max_rank - 40, y = max_rank - 5, color = "darkorange", label = "Highest emitters", hjust = 0, fontface = "italic") +
  annotate('segment', x = max_rank - 38, xend = max_rank - 26, y = max_rank - 5, yend = max_rank - 12, color = "darkorange",
           arrow = arrow(type = "closed", length = unit(0.1, "inches"))) +

  labs(
    title = "Changes in CO2 emission ranking between 1995 and 2020",
    x = "Rank in 1995",
    y = "Rank in 2020",
    color = "Change Status"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "right"
  )

```

Changes in CO2 emission ranking between 1995 and 2020

